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The Limit on AI Agents May Be the People Who Supervise Them

Count the review work before approving the next agent

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EXECUTIVE SUMMARY

Oversight may be the first limit on how many AI agents a company can run well, so count the human review work before approving another.

Every AI agent adds output. It also adds work for people who check important results, handle exceptions, and answer when the AI is wrong. Agents can produce work in seconds, but human attention is unlikely to grow at the same speed. This paper shows how to count that work before ownership becomes a name on a chart, and says where the evidence stops.

- 01 Ask one named owner to explain what the agent did, what was checked, what failed, and what changed. If the owner cannot, check the owner's hours against what was expected before narrowing, because thin records can fail the test too.
- 02 Match review to risk. Sample easy, low-risk work and require a person for high-risk or unusual cases.
- 03 Count review time as a cost in every funding choice. Often the first move is to shift reviewer time from stable, low-risk work to higher-risk work before adding more AI.
- 04 If no one can estimate how much AI work one person can supervise, start with one owner, one workflow, and a cap on live volume, and widen only after four weeks of records show room.

The limit nobody counts

You are told to add more AI agents, find more uses, and get more work done. The plan may count model cost, data, and technical skill. It may not ask a harder question: How many AI decisions can your people still review, explain, and stand behind?

The claim here is narrow. Human oversight can become the first limit a company hits as it adds AI agents, and a plan can easily leave it uncounted. The evidence supports that claim, not a stronger one. It does not show that oversight always sets the limit, and it gives no number a company can borrow. Better models and automated checks may lower the review load a great deal. The test is simple: if the owner's review hours

stay well within the time the owner has as agents or cases are added to a workflow, oversight is not the limit for that workflow, and the weekly log described at the end will show which.

The problem is newer than it looks. An agent can create output in seconds. A person still needs time and judgment to check important results, handle exceptions, and answer when the AI is wrong. Producing the work used to be the slow step, so review kept pace without anyone planning for it. Agents make output cheap, which likely makes review the slower step. That is this paper's inference, not a measured result. If agents grow faster than the human work behind them, the company gains output that no one can fully own.

McKinsey's authors expect human oversight to cap how widely companies adopt AI agents.[1] That is a forecast, not a measurement. They also say that, in their experience, a team of two to five people can already supervise 50 to 100 specialized agents running one end-to-end process.[1] The article says its lessons come from early adopters and gives no sample or method for the figure. Read it as one consultancy's early observation, not a safe benchmark. It still points to the question leaders need to answer: Where is the limit for this team and this work?

The sections that follow show how to estimate supervision, spend it by risk, extend it with tools, and share it across all of a company's AI work. Estimating comes first because the other choices depend on knowing the load. The rest can be taken in any order. The objections and what the evidence does not show come before "What to do next."

What the team runs out of

The scarce resource is human oversight: the time and judgment people use to review AI output, own decisions, and respond to failures. This paper uses oversight and supervision to mean the same thing. It has three traits.

First, it is finite. One person can own only so many decisions before ownership becomes a name on a chart. Second, it is shared. A new agent may need help from security, legal, risk, and operations. Those experts also support other AI workflows. Third, it is used whether anyone counts it or not.

That last trait makes the problem easy to miss. A company can add agents without adding the people, tools, or time needed to supervise them. Output rises at once. The oversight gap likely appears later, when a hard case arrives or a failure reaches another team.

Model quality may open the door, but human oversight can set how much AI the organization can keep live. Leaders can manage that limit once they treat it as work that can be named, measured, and funded.

Estimate how much one person can supervise

Start with the span of supervision: how much AI work one accountable person can genuinely own. Pick one unit for each workflow, such as agents, cases a week, or decisions, and keep to it. There is no single number for every workflow. The span gets smaller when errors are costly, hard to see, or hard to reverse. It can be larger when errors are easy to spot and fix.

Two simulation studies show why attention may set a span. Both are laboratory studies of military-style tasks, not business AI. In the first, one operator supervised several semi-autonomous unmanned aircraft. Some operators reached cognitive saturation, meaning too much to track at once, and changed how they managed the work. Operators given an aid that flagged only the exceptions showed signs of automation bias, which means accepting a machine's suggestion without checking it. [2] The study did not measure how performance changed as aircraft were added. In the second, 42 U.S. Navy personnel controlled 8, 12, or 16 simulated missiles while handling secondary tasks. At 16, performance on three measures and awareness of the situation fell significantly.[3] The authors proposed that an operator who is busy about 70 percent of the time is near the point where performance starts to decay.[3]

Neither study sets a limit for business AI. They show that one person's attention can become the weak point even when the machines act with some independence. The second also suggests what to measure: how much of the owner's week the work actually fills. The 70 percent figure comes from one simulation, so treat it as a reason to measure, not as a target.

Estimate the span for each workflow. Ask four questions:

- How much harm can one wrong result cause?
- How hard is an error to detect?
- How often does the agent meet a case it has not seen?
- How much time does the owner need to review exceptions and follow up?

Sometimes no one can answer them. The workflow is new, there is no record of its errors, and the owner cannot separate the agent's work from other duties. Treat that as a reason to hold expansion, not to stop work already approved. Start at the

narrowest span: one owner, one workflow, a person checking every case or a large sample, and a cap on live volume. For a workflow that is already live, that means no new agents or cases until the records exist. Widen only after four weeks of records show the owner has room. A weak estimate narrows the span. A missing estimate keeps it at the starting point until the records exist.

Then use a weekly test: Can the named owner explain what the agent did, what was checked, what failed, and what changed? If the honest answer is no, the team may have taken on more than it can supervise.

A failed test has two possible causes. The owner may be out of time. The records may also be too thin, so even a well-staffed owner cannot say what happened. Poor records are a false alarm for overload, but they are still a real gap, because an owner cannot supervise what no one wrote down. Ask the owner two follow-up questions. How many hours did review and follow-up take this week, compared with the hours the owner expected? And were the records there to answer from? If hours ran over, or review was skipped, narrow the span. If hours were within what the owner expected but the records were missing, fix the records first and add no agents in the meantime.

A pass proves less than it seems. An owner can explain a process that no one has checked closely. Once a month, ask a reviewer outside the workflow to test one of the owner's answers against the logs. If the logs do not support the answer, count the test as failed.

Review by risk, not every action

A person need not read every output. That would place all AI work back at human speed. The goal is to use attention where an error matters most.

Low-risk, easy-to-check work may need sampling. High-risk work may need a person to approve each case. New or uncertain work may need a tighter review until the team understands its typical failures. The rules should say what is sampled, what always gets a human check, what triggers a deeper review, and who can stop the workflow.

McKinsey describes people moving “above the loop”: they set policy, watch unusual cases, and change how involved they are instead of checking every action.^[1] This approach can stretch a fixed amount of attention. It works only when the review rules are written and the exceptions reach someone who can act. “The Exception Path Is the Real Automated Workflow” covers how to design that route.

Use automation to extend reviewers

Technology can help people supervise more work. McKinsey describes critic agents that challenge output, guardrail agents that enforce policy, and compliance agents that watch rules.[1] Deloitte says oversight may include agents guarding other agents that are then guarded by humans.[4]

Public standards add practical controls. The NIST Generative AI Profile suggests ongoing monitoring, red-team exercises, independent evaluation, incident response, and structured feedback.[5] The NIST AI RMF Core organizes AI risk work into four functions, govern, map, measure, and manage, and treats monitoring as ongoing.[6]

These controls can find common errors and send unusual cases to people. They do not remove human ownership. A critic agent can miss a problem. A guardrail can be too broad or too weak. The automated checks also need tests, monitoring, and someone who can change or stop them.

Use automation as a multiplier for a human team. Count the work needed to maintain that automation as part of the review load.

Treat oversight as shared capacity

Every live AI workflow draws on a shared group of reviewers and experts. That makes oversight a choice across the whole portfolio of AI work. A new agent may look useful on its own while still taking time away from a more important workflow.

Each funding request should state its expected review load. It should name the owner and the type of review. It should also list the experts needed, the main warning signs, and the signals that would pause or stop the work. The portfolio owner, the person who funds and compares the AI work, can then compare that demand with the people who are actually available. A request that does not state its review load waits until it does.

When the shared group of reviewers is fully booked, the first move should often be reallocation. A stable, low-risk workflow may need less review than it did at launch. Those hours can move to a new or high-risk workflow. This may support more useful AI without hiring another reviewer, though no source here measures how often it works.

More model features do not solve a shortage of supervision. Once people are the limit, leaders should compare the next model expense with monitoring, evaluation,

workflow design, and reviewer time.

Watch for three warning signs

The first warning is ownership in name only. The workflow has an owner on paper, but that person cannot say what it did this week or which exceptions remain open.

EXHIBIT 1

Signs that supervision is too thin, and the response

Owner in name only

The owner cannot explain what the agent did this week.

Rubber-stamp review

A person approves work without enough time or evidence to inspect it.

Surprise incidents

Failures appear in workflows no one was watching closely.

The response

Narrow the work, pause new uses, or add review before adding AI.

The second warning is rubber-stamp review. A person approves output without enough time or evidence to inspect it. The approval creates a record of control that did not really happen. A review of mostly laboratory studies found that people monitor automation less closely when other tasks compete for their attention and when the automation is consistently reliable.[7] A reviewer with a full queue and a reliable agent is where this is most likely.

The third warning is surprise. A failure appears in a workflow that no one was watching closely. The incident may be treated as a model problem even when the deeper cause was thin supervision.

When these signs appear, slow something down. Reduce the number of live cases, narrow the agent's role, pause new uses, or add review before adding more AI. A better model may help later, but it does not replace the missing owner. These signs are what thin supervision can look like. No source here says how often each one occurs.

Keep the people who can catch the error

Oversight comes from specific people who know the work. They can tell when an answer looks polished but is wrong for the case. That judgment takes time to build.

This paper's inference is that a company can weaken its own controls when it cuts experienced staff to fund AI. The work may appear to move faster, but fewer people remain who can test an unusual result or understand its effect on customers, employees, or operations. The company adds AI at the same time that it removes the people who can safely own it. "The AI Replacement Boomerang Is a Governance Failure" reports what analysts forecast about companies that cut staff for AI.

Reviewer time belongs in the workforce plan. Leaders should identify which skills each workflow needs, how many trained owners are available, and how new owners will learn. This is part of the cost of running AI, not spare effort that appears after launch.

Objections, and what the evidence does not show

Better models and automated checks will remove the limit. They may. Critic and guardrail agents are real, and the span may widen as they improve. This paper does not say the limit is permanent. It says to count instead of assume, and the weekly log shows whether review hours stay well within the owner's available time as volume rises. The automation also needs upkeep, which counts as review work.

Sampling will miss a serious failure. Sometimes it will. That is why sampling fits only low-risk work whose errors are cheap and easy to catch, and why costly or hidden errors get a person on each case.

Money, data, and skills limit AI too. They do. Oversight may bind first in some workflows and not in others. Where errors are cheap and easy to fix, review stays light and this paper's concern does not apply.

The evidence is thin. It is. The McKinsey figure is an early observation with no stated method.[1] The simulations are military tasks and the review is mostly laboratory work, so they support counting attention but do not tell a company its number.[2][3] [7] The return on spending more for oversight is also not well measured. McKinsey, Deloitte, and NIST describe useful practices, but none gives a formula for the right staffing level.[1][4][5][6] Start with a cautious estimate. Measure the real load, missed issues, and response time. Then change the span with evidence.

What to do next

Choose one live AI workflow and build a simple supervision record:

EXHIBIT 2

A simple supervision record

1. Name the owner

One person answers for the AI agent's results.

2. Set the review rule

Say what is checked, sampled, or sent for deeper review.

3. Count the time

Log review hours beside live volume each week.

4. Name the warning signs

Write the events that trigger a deeper review.

5. Name the experts

List who the workflow may need for hard cases.

6. Set a stopping point

Know when to narrow, pause, or stop the AI workflow.

- Name the person who owns the result.
- List the output that receives a human check and the output that is sampled.
- Record the warning signs that trigger a deeper review.
- Estimate the owner's weekly review and follow-up time.
- Name the experts the workflow may need for hard cases.
- Set the point at which the team will narrow, pause, or stop the work.

The workflow's owner writes the record, with the portfolio owner who funds the work. It is a one-page working document, kept current and reviewed whenever the workflow changes. As a planning estimate, not a measured figure, expect a few hours to write the first one and a few minutes each week to log the owner's review time. The larger cost is the reviewer time the record exposes, which is the point of writing it.

Each week, log two numbers: the owner's review and follow-up hours, and the live volume behind them, counted in the unit chosen for the workflow. Review the record after four weeks. Compare the expected load with the time people actually spent.

Check whether exceptions reached the right person and whether the owner can explain what changed.

Then decide the span. If the owner had room and the records were clear, widen the span one step, meaning one more agent or one more group of cases. When that step is taken, compare hours before and after. Hours that stay well within the time the owner has as volume rises mean oversight is not yet the limit for that workflow. Hours that climb toward all the time the owner has mean it is, and the span should narrow before another agent goes live. If time ran over or review slipped in the first four weeks, narrow the span at once, without waiting for the four-week review. The four weeks, the monthly outside check, and the one-step rule are this paper's design, not research findings.

The choice leaders control

More agents can raise output for a while. The harder test is whether named people can still explain the work, catch important errors, and act when something fails.

Count that human work before approving the next agent. Fund it as part of the cost of running each agent. Move it toward the highest risk. Keep the experienced people who supply it. Oversight may not be your limit, but you will not know until you count.

Notes and sources

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About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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